SIMATS ENGINEERING

ASSESSMENT TOOL 2-Pseudocode Writing Task

COURSE NAME: R PROGRAMMING **COURSE CODE: ITA0402**

COURSE OUTCOME COVERED

CO1: Apply R programming fundamentals including data structures, vectors, matrices, arrays, and class types to perform mathematical operations and manage data effectively. Identify and implement appropriate control structures to solve a particular programming problem
(Bloom Level -BL3- BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 Questions **Total Marks: 100**

Code of Conduct:

I RAKSHITH.R.R(192324136) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Rakshith RR

Instructions to students: Write structured pseudocode (not R syntax) for each scenario. Clearly show inputs, data structures used, control structures (decision, loop), processing steps, and outputs. Marks: 10 per question.

Q1. Campus Canteen Billing (BL3: Vectors)
The campus canteen stores item prices in a numeric vector price and the quantity ordered in a vector qty for one customer. Write pseudocode to compute the bill for each item, the total bill, and apply a 10% discount if the total exceeds ₹500. Display the final payable amount.
Expected concepts: vector arithmetic, element-wise multiplication, sum, if-else.

Q2. Student Marks Analyser (BL3: Vectors, Control Structures)
A teacher stores the marks of 40 students in a vector. Write pseudocode to find (a) the class average, (b) the highest and lowest mark, and (c) the count of students who scored below 40 (fail). Print "Class Performance: Good" if more than 80% have passed, otherwise "Needs Improvement".
Expected concepts: vector functions, logical indexing, counting with condition, if-else.

Q3. Hostel Room Allocation Matrix (BL3: Matrices)
A hostel has 4 floors and 5 rooms per floor. A 4×5 matrix stores 1 if a room is occupied and 0 if vacant. Write pseudocode to display (a) the number of vacant rooms on each floor, (b) the total vacant rooms in the hostel, and (c) the floor with the maximum vacancies.
Expected concepts: matrix creation, row-wise sums, nested loops or apply-style logic, finding maximum.

Q4. Monthly Electricity Bill Calculator (BL3: Control Structures)
An electricity board charges ₹3/unit for the first 100 units, ₹5/unit for the next 100 units, and ₹8/unit beyond 200 units. Given the units consumed by a household, write pseudocode to compute the total bill. Add a fixed charge of ₹50 and print the bill summary.
Expected concepts: nested if-else / if-else-if ladder, slab-based calculation.

Q5. Sales Report Using Matrix Operations (BL4: Matrices, Analysis)
A shop records the sales of 3 products over 4 weeks in a 3×4 matrix. Write pseudocode to (a) compute the total sales per product, (b) compute the weekly total sales, (c) identify the best-performing product and the best week, and (d) transpose the matrix so weeks appear as rows.
Expected concepts: matrix row/column sums, which.max-type logic, transpose, analysis of results.

Q6. Library Book Management Using Lists (BL3: Class Types and Data Structures)
A library stores details of a book (title, author, year, availability as TRUE/FALSE) in a list. Write pseudocode to (a) create a list of 5 such books, (b) display all books published after 2015, and (c) update availability to FALSE when a user borrows a book by title. Show a message if the title is not found.
Expected concepts: list of lists, loop traversal, condition checking, element update.

Q7. Temperature Data Cleaning (BL4: Vectors, Loops, Analysis)
A weather station records 30 days of temperature readings in a vector, but some entries are NA (missing) due to sensor failure. Write pseudocode to (a) count the missing values, (b) replace each missing value with the mean of the valid readings, and (c) classify each day as "Cold" (<20°C), "Moderate" (20–30°C), or "Hot" (>30°C), storing the result in a new vector.
Expected concepts: NA handling, mean of valid data, loop with if-else ladder, building a new vector.

Q8. Exam Seating Using 3D Arrays (BL4: Arrays)
An examination hall has 3 blocks, each with 4 rows and 5 columns of seats. A 3D array stores the register number of the student seated in each position (0 for an empty seat). Write pseudocode to (a) count the occupied seats in each block, (b) find the block with the highest occupancy, and (c) display the register number at Block 2, Row 3, Column 4.
Expected concepts: 3D array indexing, nested loops across dimensions, counting non-zero elements.

Q9. Employee Data Frame Processing (BL4: Data Frames and Class Types)
An HR department maintains a data frame with columns EmpID, Name, Department, and Salary. Write pseudocode to (a) display employees from the "IT" department earning above ₹50,000, (b) compute the average salary per department, and (c) give a 10% raise to employees earning below ₹30,000 and add a new column NewSalary.
Expected concepts: data frame filtering, grouping logic, conditional update, adding a column.

Q10. Prime Number and Series Generator for a Maths Club (BL3–BL4: Loops, Vectors)
A maths club needs a program that accepts a number N from the user. Write pseudocode to (a) generate all prime numbers up to N and store them in a vector, (b) display the sum and count of those primes, and (c) print the first N terms of the Fibonacci series, also stored in a vector. Validate that N is a positive integer before proceeding.
Expected concepts: input validation, nested loops with a flag, while/for loops, vector appending, algorithm design.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

ITA0402

STATISTICS WITH R PROGRAMMING

ASSESSMENT TOOL

-02-

PSEUDO CODE WRITING TASK



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11. Campus Canteen Billing

Aim:-

To calculate the bill for each item, find the total bill, and apply a 10% discount when the total exceeds ₹500.

pseudocode:-

```
INPUT price vector
INPUT qty vector

SET itemBill = price x qty element-wise
SET totalBill = SUM(itemBill)
IF totalBill > 500 THEN
    SET discount = totalBill x 10 / 100
ELSE
    SET discount = 0
END IF
SET finalAmount = totalBill - discount
Display itemBill
DISPLAY totalBill
DISPLAY DISCOUNT
DISPLAY finalAmount.
END.
```

Example:-

Price = [100, 200, 50]

Quantity = [2, 1, 3]

Item Bills = [200, 200, 150]

Total Bill = ₹550

Discount = ₹55

Final Amount = ₹495.

Q2. Student Marks Analyzer:-

Aim:-

To analyze the marks of 40 students by calculating the class average, highest mark, lowest mark, number of failures, and overall class performance.

Pseudocode:-

```
INPUT marks vector containing 40 student marks.
SET average = SUM(marks)/40
SET highest = MAX(marks)
SET lowest = MIN(marks)
SET failCount = 0
FOR each mark in marks DO
    IF mark < 40 THEN
        INCREMENT failCount by 1
```

```
    END IF
END FOR
    SET PassCount = 40 - FailCount
    SET PassPerCentage = (PassCount/40) X100
    IF PassPercentage > 80 THEN
        SET performance = "Class Performance: Good"
    ELSE
        SET performance = "Class Performance: Needs
            Improvement"
    END IF
    DISPLAY average
    DISPLAY Highest
    DISPLAY Lowest
    DISPLAY FailCount
    DISPLAY passPerCentage
    DISPLAY performance.
    END.
```

Example:-

Average = 68.5

Highest = 98

Lowest = 25

Failed students = 4

Pass Percentage = 90%

Class Performance : Good.

34. Monthly Electricity Bill Calculator:-

AIM:-

To calculate the electricity bill using different unit-based slabs and add a fixed charge of ₹50.

pseudocode:-

```
INPUT units
IF units <=100 THEN
    SET energyCharge =units*3
ELSE IF units <=200 THEN
    SET energyCharge =(100*3)+((units-100)*5)
ELSE
    SET energyCharge =(100*3)+(100*5)+((units-200)*8).
END IF.
SET fixedCharge =50
SET totalBill= energyCharge + fixedCharge.
DISPLAY units
DISPLAY energyCharge
DISPLAY fixedCharge
DISPLAY totalBill.
END.
```

Example:-

Units = 250

Then :-

First 100 = 100x3 = ₹300

Next 100 = 100x5 = ₹500

Remaining 50 = 50x8 = ₹400.

Energy Charge = ₹1200

Fixed Charge = ₹50.

Total Bill = ₹1250.

9b. Library Book Management Using Lists :-

Aim :-

To manage library book details using a list, display books published after 2015, and update the availability of a borrowed book.

Pseudocode :-

```
BEGIN
CREATE empty library_list
FOR i=1 TO 5 DO
    INPUT title
    INPUT author
    INPUT year
    INPUT availability
```

```
CREATE book list containing :--
        title
        author
        year
        availability
    ADD book to library list
END FOR
DISPLAY "Books published after 2015"
FOR each book in library DO
    IF book.year > 2015 THEN
        DISPLAY book.title
        DISPLAY book.author
        DISPLAY book.year
        DISPLAY book.availability
    END IF
END FOR
INPUT borrowTitle.
SET found = FALSE
FOR each book in library DO
    IF book.title = borrowTITLE THEN
        SET book.availability = FALSE
        SET found = TRUE
```

```
        DISPLAY "Book borrowed Successfully"
        BREAK
    ENDIF
ENDFOR
If found = FALSE THEN
    DISPLAY "Book title not found"
ENDIF
END
```

Example:-

Book 1:-
- Title = Python Programming
- Author = John
- Year = 2020
- Availability = TRUE.

Book 2:-
- Title = C programming
- Author = Dennis
- Year = 2010
- Availability = TRUE.

07. Temperature Data Cleaning :-

Aim :-

To identify missing temperature readings, replace missing values with the mean temperature, and classify each day's temperature.

Pseudo Code :-

```
BEGIN
INPUT temperature vector containing 30 readings.
SET missingCount = 0
FOR each temperature in temperature vector DO
    IF temperature is missing THEN
        INCREMENT missingCount by 1
    END IF
END FOR
SET validTemperatures = temperatures excluding missing values
SET meanTemperature = MEAN(validTemperatures)
FOR each position i from 1 to 30 DO
    IF temperature[i] is missing THEN.
        SET temperature[i] = meanTemperature
    <-> ENDIF.
```

```
END FOR

CREATE empty Classification vector

FOR each Temperature in temperature vector DO

    IF Temperature < 20 THEN

        APP :"Cold" to classification vector

    ELSEIF temperature <= 30 THEN

        APP "Moderate" to classification vector

    ENDIF

END FOR

DISPLAY missing count

DISPLAY mean Temperature

DISPLAY cleaned Temperature vector

DISPLAY Classification vector

END
```

Example:-

Input:- [18, 5, NA, 32, 15]

Valid values:[18, 25, 32, 15

Mean: 22.5

After Cleaning:[18, 25, 22.5, 32, 15]

Classification:- [Cold, Moderate, Moderate, Hot, Cold]